

# Efficiency and Equity

## ▼ What is efficiency?

Producing the goods that society wants at the lowest possible cost

Maximising production for the lowest possible cost

## ▼ What is pareto efficiency?

The situation where trying to make someone better off would also make someone else worse off

Happens at market equilibrium

## ▼ What are the types of efficiency?

Productive efficiency

Where goods and services are produced at the lowest possible cost

Allocative efficiency

Production reflects consumer preferences

Dynamic efficiency

Occurs when new technology and innovations are adopted over time

## ▼ What is consumer surplus?

The difference between what a consumer is prepared to pay and what they actually pay in the market

## ▼ What is marginal benefit?

The extra benefit from consuming one extra unit of good or service

The more you consume, the less benefit you gain from each extra unit

## ▼ What is producer surplus?

The difference between what a producer is willing to receive and what they actually receive in the market

Usually what they are willing to receive will be the marginal cost of production

Producer surplus = Total revenue - Total cost

## ▼ What is marginal cost?

The extra opportunity cost of producing one more unit of a good or service

▼ **What is total surplus?**

The measure of the net benefits to society from the production and consumption of the good or service

The aim for society is to maximise total surplus in every market

$$CS + PS = TS$$

$$TS = TB - TC$$

Maximised at equilibrium where  $MC = MB$

If the market is producing to the left of the equilibrium where total surplus is not maximised, and the MB exceeds MC, total surplus and resource allocation efficiency could be increased by increasing production from producers

If the market is producing to the right of the equilibrium where total surplus is not maximised, and the MC exceeds MB, producers and society are losing money and resources by producing in a surplus

▼ **What is deadweight loss?**

A loss in total surplus that is avoidable

Not producing at optimal output

▼ **What is equity?**

Refers to the equal distribution of resources across the society

▼ **What are the types of equity?**

Vertical equity

Equity between high and low income levels

Horizontal equity

To provide everyone with the same opportunity to succeed and earn a high income i.e. equal opportunity

▼ **What are policy options to promote equity?**

Market restrictions

Price ceilings/floor

Taxes

Subsidies

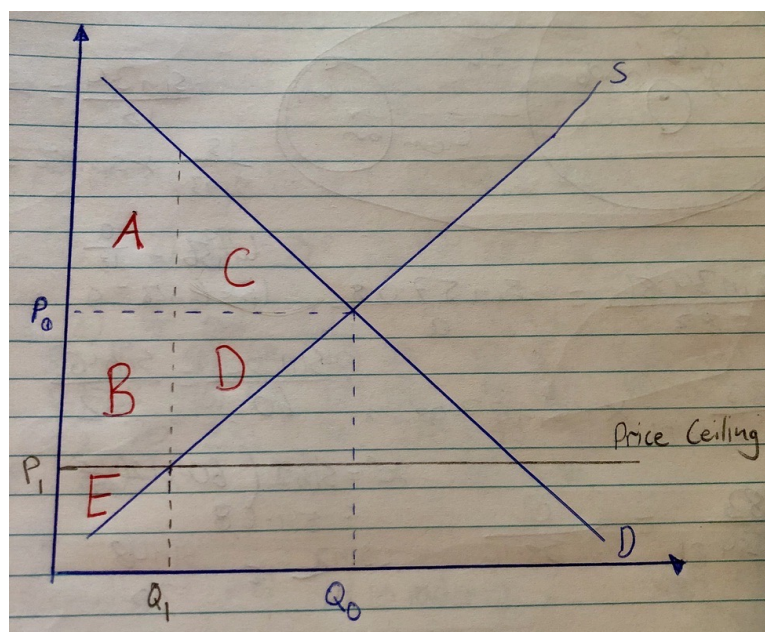
Social security

Enforcement of private property rights

▼ **Describe the impact of price ceilings on efficiency and equity**

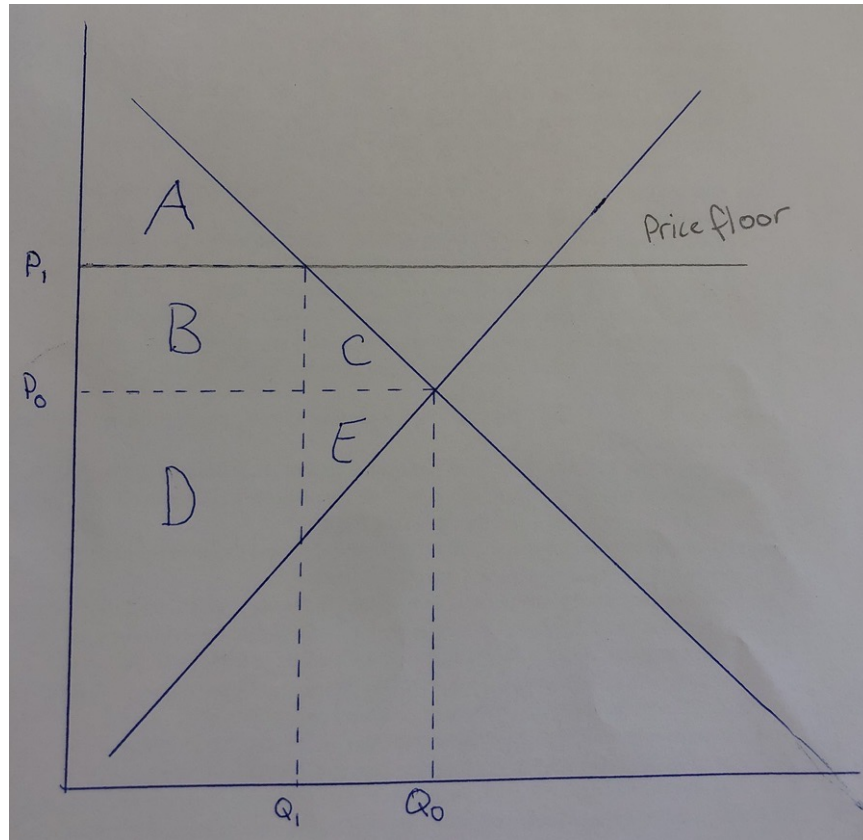
Where the government imposes a maximum price within the market

The price ceiling is drawn below the equilibrium price



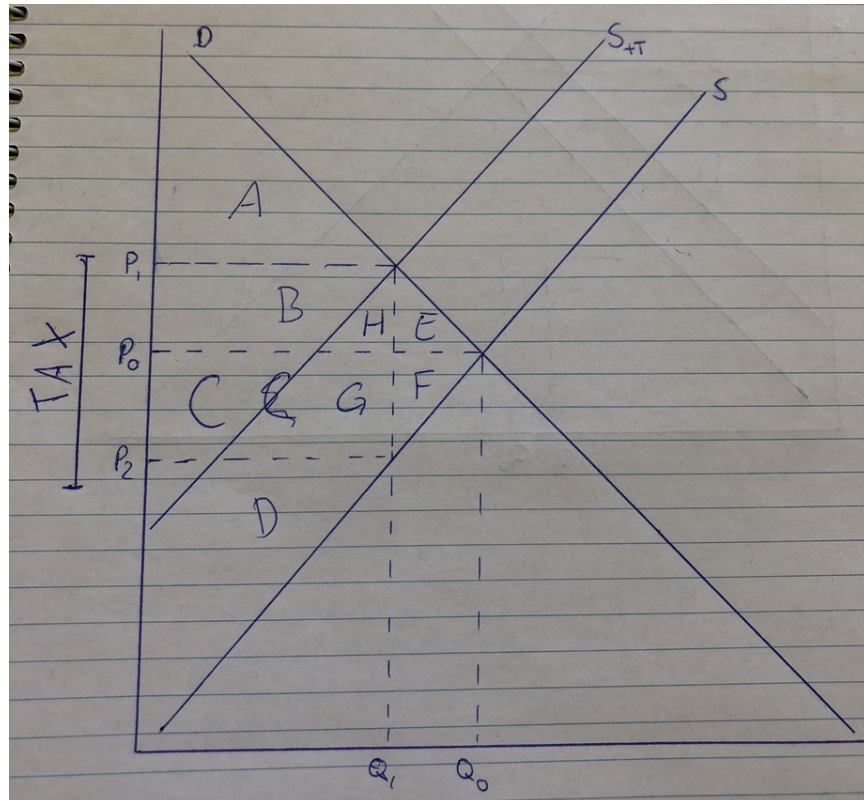
Before the price ceiling when equilibrium was at  $P_0$  and  $Q_0$ , the consumer surplus was originally at areas A and C and producer surplus was at a B, D and E. After the enforcement of the price ceiling the producer must produce at  $P_1$  and therefore at  $Q_1$ , to maximise production and thus profits. This alters CS by removing area C and adding area B. PS also changes by removing areas B and D, reducing PS to only area E. Now, CS is greater than it was before **indicating a greater benefit of the price ceiling on the consumer**, not the producer. This means that areas C and D are now dead weight loss as they are not included in the potential total surplus of the market, indicating an avoidable loss in market efficiency.

▼ **Describe the impact of price floors on efficiency and equity**



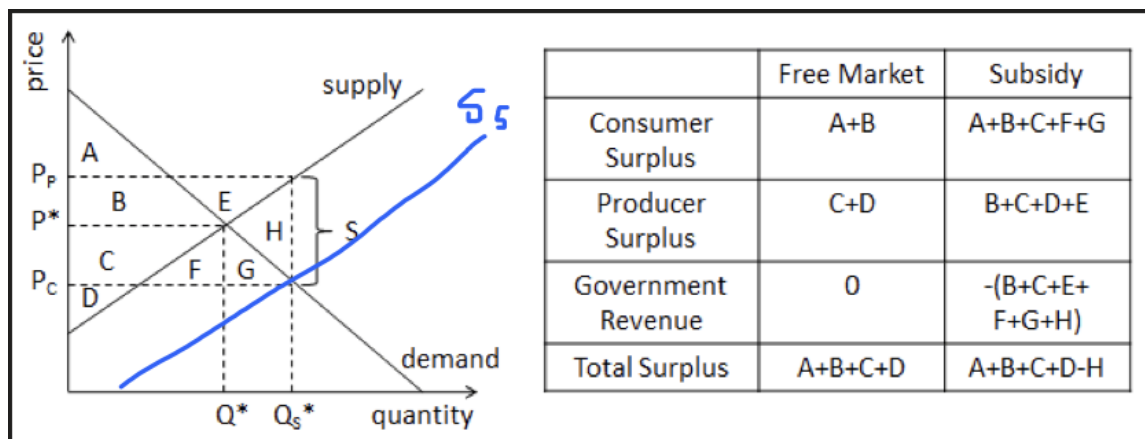
Before the price floor, with the objective of preventing prices of being too low for producers, the equilibrium price and quantity was at  $P_0$  and  $Q_0$ . At this free market the CS was represented by areas A, B and C and the PS in areas D and E. After the price floor was implemented, producers **decreased quantity supplied** to  $Q_1$  and charged **a higher subsequent price** of  $P_1$ . This new clearance price and quantity decreases CS to only area A and increases PS to areas B and D, thus benefiting producers rather than consumers. Regions C and E are regarded as DWL as they are not included in the potential total surplus of the market and represent market inefficiency because of the price floor.

▼ **Describe the impact of taxes on efficiency and equity**



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▼ Describe the impact of subsidies on efficiency and equity



Deadweight loss in the market will only happen if the market is already mostly inefficient and the government wants more out of that industry and lower prices. The deadweight loss is created because [the price the government pays for the subsidy is greater than the benefit to society](#). The opportunity cost of spending money on a subsidy on an inefficient market is greater than the opportunity cost of spending the same amount of money on an efficient industry or other areas the government needs.

#### ▼ Outline the trade off between efficiency and equity

- Efficiency emphasises maximum economic benefit
- Equity emphasises equal treatment/opportunity
- Government policies options to improve equity often can cause inefficiencies
- Depending on the type of market failure, targeting inefficiency can also improve equity